Arika Khor

Interest in Robotics, Network Engineering, and Real Time Facial Recognition.

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Education

Expected B.S. Fall 2026 , M.S. 2027

The University of Oklahoma
B.S. / M.S. Computer Science

Experience

June 2025 - Current

Construction Innovation and Analytics Lab
Researcher

- Implemented an autonomous 4-wheel differential drive robot navigating on a known dynamic construction site.
- Used Nav2, ROS2, and multi-sensor fusion SLAM with a zed2i (depth camera) and Unitree LIDAR.

Aug 2025 - Current

Autonomous Interactive Robotics Lab
Researcher

- Passive Reinforcement learning visual-only autonomous navigation with Nvida Isaac Sim and Isaac Lab.
- Implemented a **smart intersection** using a traffic controller with Multi-agent reinforcement learning robots

Sept 2022 - July 2024

Singapore Armed Forces
Team Lead for Curriculum Development

- Prediction based data analysis on Cadet scores using pandas.
- Led a project to develop code to migrate from obsolete excel sheets that were used in trainee scoring.

Jan 2020 - Dec 2020

GameDuchy *Game Development Company*
Assistant Community Relations Manager

- Planned, organised and executed regular in-game community wide events, working with various departments to successfully implement these events.
- Designed and implemented custom chat bot to obtain item data.

Leadership Roles

Aug 2024 - Current

Sooner Competitive Robotics
Member (24 - 25), Treasurer (25 - 26)

- Learnt about HSV color thresholding for object detection and localization.
- Setup and managed Docker environments. Overhauled UI to add a live remote logging viewer, configuration manager, and overhead map. Overhauled data transfer from web socket to WebRTC request.

Aug 2016 - May 2022

Scouting America
Various Leadership Positions

- **Eagle Scout Project:** Designed and built several weather radio antennas by organizing a large group to mass-produce them. They were then given to the city's emergency coordinator to distribute to public service buildings. This project was covered in the normantranscript (article **here**).

Aug 2024 - Current

Sooner Competitive Programming Team
Team Capitan (24-26, Team Advisor (26-27)

- Attended 2025 and 2025 ICPC Programming competitions and hosted team practices solving kattis problems.

Projects

2025 - Current

Designing a Control System for Autonomous Cognition of Latent Hazards in Active Construction Sites
Paper, Abstract accepted to Smart and Sustainable Built Environment Journal

2025 Hacklahoma

Facial-Net Classifier
Facial Recognition Project

- Fully local real time Facial Recognition suite to raise awareness about academia and policy making.
- Captures live video then detects and matches faces in a crowd attempting to either match or generate a new face with mongodb.
- Used quartz 4 and obsidan for documentation.
- UI to merge/delete/assign saved faces using Streamlit and flask.

Technical & Soft Skills

- Programming: Advanced Java (Spring,Thymeleaf), C++, C#, Java Script, Python, Bash
- Web: LaTeX, HTML5, CSS3, Bootstrap, Azure, AWS, SQL, MonogDB, Bun JS, React, WebRTC Concepts
- Tools & Concepts : ROS2, Pub/Sub Architecture, Docker, Jira, Jenkins , Git, Microsoft Office Suite, Agile, Scrum, tailscale.
- Soft Skills: Leadership, Interdisciplinary Teamwork, Communication, Problem Solving, Project management.
- *OS and Machines*: Debian, Windows, Jetson AGX Orin (Jetpack 6.2)